

# Order Crossover (OX)

|          | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |                           |
|----------|---|---|---|---|---|---|---|---|---------------------------|
| Parent A | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Segment: positions [2, 4] |
| Parent B | 2 | 6 | 4 | 0 | 5 | 7 | 1 | 3 |                           |

Step 1 -- Copy segment from Parent A

|       |  |  |   |   |   |  |  |  |
|-------|--|--|---|---|---|--|--|--|
| Child |  |  | 2 | 3 | 4 |  |  |  |
|-------|--|--|---|---|---|--|--|--|

Step 2 -- Scan B from pos 5 (wrapping), skip {2,3,4}: collect [7,1,6,0,5]

Step 3 -- Fill empty positions from pos 5 onward with collected values

|        |   |   |   |   |   |   |   |   |
|--------|---|---|---|---|---|---|---|---|
| Result | 0 | 5 | 2 | 3 | 4 | 7 | 1 | 6 |
|--------|---|---|---|---|---|---|---|---|

From Parent A From B (relative order)